A Sufficient Statistic for Algorithmic Feedback: What Prices Reveal About Trader Groups, and Who Needs the Rest

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Abstract

In a market with algorithmic feedback traders, the feedback kernel—the response of total feedback flow to past prices—is a sufficient statistic for aggregate dynamics. For any insider and causal dealer rule, populations with the same kernel generate identical prices, profits, and equilibria, and public prices identify nothing finer. Learning group composition costs the known $(\Delta\eta)^{-(4K-2)}$ mixture rate, which survives the closed loop and holds in an inventory equilibrium. Position limits introduce a second layer: a capacity measure over memory and leverage. Calm markets reveal only the linear kernel, carrying zero information about proximity to hard limits. Stress large enough to reach limits reveals the capacity measure exactly. In a pre-registered out-of-sample test on Taiwan tagged exchange flows (2019–2023), predictive gains pass the registered criterion, but pre-specified diagnostics attribute them to prices revealing past flow rather than feedback. Core statements are machine-checked in Lean 4.

AI-use disclosure. Claude, ChatGPT/Codex and Gemini provided material assistance with research development, code, numerical checks, manuscript drafting and internal critique, including the restructuring of this core version. ChatGPT/Codex additionally developed and checked Lean proofs. Human authors remain responsible for verifying the content and its scholarly contribution before submission. AI checks do not constitute independent review, and no independent human proof check has been completed.

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Relation to the full working paper. This is a shortened version, organized around one result. Every result stated here also appears in the full working paper ("What Prices Cannot Tell a Market Supervisor", 5 November 2026), with the same proof. The exceptions are Proposition 3.2 and all of Section 7, which are new to this version and have not appeared elsewhere. Results not needed for the main argument are omitted, among them account-level capital, general memory, spectral blindness and every empirical test. Section 8 says where they are.

Keywords: algorithmic trading, feedback trading, sufficient statistics, aggregation, identification, market microstructure, supervision, disclosure.

JEL: C32, D83, G14, G18.

Code and proofs. All code, Lean proofs and result files are public at https://github.com/Pratikn03/Fianince.

1 Introduction

Much of today's trading is done by programs that read recent prices, decide what to trade, and by trading move the prices they read. Trend-following funds, execution algorithms, market-making programs and, more recently, AI trading agents all work this way. When such a market misbehaves, supervisors ask who was trading and how fast those traders react. This paper asks a prior question: what does a market's price record reveal about its population of feedback traders, and for which decisions is that enough?

The answer in one sentence. Prices reveal the market's *feedback kernel*: the response, m periods later, of total feedback flow to a unit move in the price signal. That kernel is all that prices identify, and it is all that anyone trading on public information needs. Group composition, meaning who the groups are and how strongly each trades, matters only for actions aimed at a particular group, and learning it from prices has a known and steep price.

The model. The market has a price and K groups of traders. Each group keeps an exponentially weighted average of past prices, with memory rate r_i, and trades in proportion to it with weight w_i. A desk that tracks a trend with a Kalman filter, or that learns by regularized online gradient descent, behaves exactly this way (Lemma 2.1). Total feedback flow is then a convolution of past prices with the kernel

$$k_m = \sum_i w_i r_i^m, \qquad m = 0, 1, 2, \ldots,$$

the moments of the *merged measure* M, which puts weight w_i at rate r_i. The paper rests on three results about this one object.

(1) *The kernel is sufficient* (Theorem 3.1 and Proposition 3.2). Add a value, noise traders, an insider and a dealer, each following an arbitrary causal rule, which covers Kyle, inventory-averse, learning and misspecified dealers. Two trader populations with the same kernel then produce the same price and order-flow paths, the same insider trades and profits and the same dealer inventories, realization by realization. The two games have the same strategy sets and payoffs, and hence the same equilibria. The proof needs only that aggregate feedback flow is a fixed linear function of past public signals, so the result holds for any linear trader population, not only exponential memories.

(2) *The kernel is all that prices identify* (Theorem 3.3). Two populations give identical price responses to every trade sequence if and only if their merged measures coincide.

Groups with the same memory cannot be separated however long the record. Groups with different memories can be separated in principle.

(3) *Learning more has a known price* (Theorems 5.1, 5.2 and 5.4). To separate K groups whose memories lie within $\Delta\eta$ of each other takes of order $(\Delta\eta)^{-(4K-2)}$ experiments, and Prony's method attains this order. Waiting instead of trading does not help: passive observation needs at least the same order of observations.

The results survive equilibrium. In the base model the price impact and the persistence of mispricing are primitives. Section 4 derives both from competitive dealers who dislike holding inventory. The equilibrium price is then the same closed loop, path by path, with effective impact and persistence and with the memory rates unchanged (Theorem 4.2). So the kernel remains the sufficient statistic, and the cost of going beyond it keeps its order. The supplement adds a strategic insider with short-lived information. It also shows that, with positive feedback, liquidity disappears beyond a threshold in dealer risk aversion that falls with the feedback intensity.

What the kernel misses: limits. Real feedback traders have position limits, margin and risk budgets. Section 7 gives a two-layer theory for traders whose positions saturate.

- The sufficient object becomes a capacity-weighted measure over memory and leverage, of which the kernel is one linear functional (Theorem 7.1).
- Calm-market prices reveal the kernel and nothing more. Under hard limits, two populations with the same kernel give identical prices until some trader reaches its limit (Theorem 7.2). Calm data therefore carry no information about how close traders are to their limits, and Gaussian data carry exponentially little. Under smooth limits, the information grows like the sixth power of the probe (Proposition 7.3).
- Stress reveals everything that matters. Responses to probes of all sizes identify the net capacity at every memory and leverage level, which is exactly the equivalence class (Theorem 7.4).
- The results extend to traders who deleverage or liquidate outside their bands (Proposition 7.5). An informed dealer who prices next-period flow leaks leverage information into calm prices, but only at the scale of the probability of a breach (Proposition 7.6).

Supervision splits in two. A market-wide action, such as a uniform cut in activity to meet a ceiling on the feedback budget, depends on the population only through the kernel. A targeted action, such as halting the largest group, does not. Two populations whose

price responses differ only by order $(\Delta\eta)^3$ can need different halts (Proposition 6.1). For such decisions, flow reported separately by trader class turns each group into an ordinary regression, with no penalty as the groups' memories approach each other (Proposition 6.2). It cuts the number of experiments needed by a factor of order $(\Delta\eta)^{-(4K-2)}$ (Theorem 6.3).

1.1 What is known and what is new

The central idea of this paper, that an aggregate is determined by a distribution of individual coefficients and not by the individuals, is old. We state its precedents first, because they fix what we can claim.

Aggregation of dynamic models. Granger (1980) showed that aggregating many first-order autoregressions with different coefficients produces a process whose dynamics are governed by the distribution of the coefficients; with enough mass near one the aggregate has long memory. Zaffaroni (2004) develops the theory for large economies. Recovering the coefficient distribution from aggregate data is known as the disaggregation problem (Celov et al., 2008). Our merged measure M plays the role of that coefficient distribution, and Theorem 3.3(i) is the corresponding identification statement for a closed feedback loop.

Sufficient statistics for aggregate dynamics. In macroeconomics, Auclert et al. (2024) show that the matrix of intertemporal marginal propensities to consume is a sufficient statistic for the output response to fiscal policy, whatever the household heterogeneity behind it. The kernel k plays the same role for a market with feedback traders. It is the response of aggregate demand to past prices, and it summarizes every heterogeneity that matters for prices.

Mixture estimation. The order $(\Delta\eta)^{-(4K-2)}$ is the known minimax rate for finite mixtures with K near-colliding components (Heinrich and Kahn, 2018; Wu and Yang, 2020), and super-resolution theory gives the same conditioning for clustered exponentials (Moitra, 2015; Batenkov et al., 2021; Katz et al., 2023). We do not claim the exponent.

Heterogeneous horizons under impact. Leclère and Rosenbaum (2026) study a linear mean-field model in which agents with heterogeneous forecast horizons trade under market impact, and in which impact depends on aggregate positions rather than on trader identities. Their object is equilibrium and price regularity, not what an observer can identify or what supervision needs.

Feedback trading and insiders. Positive-feedback traders and their effect on prices go back to De Long et al. (1990). Barberis et al. (2015, 2018) model extrapolators with exponentially weighted memories, and Brock and Hommes (1998) model heterogeneous rules. The insiders and dealers covered by Section 3 include those of Kyle (1985) and of inventory models in the tradition of Ho and Stoll (1981) and Grossman and Miller (1988).

What is new. Five things, each of which uses the market setting.

(a) *Sufficiency inside a strategic market.* The aggregation literature concerns the statistical law of an aggregate series. Theorem 3.1 is a statement about games: it holds path by path for every insider strategy and every causal dealer rule, so it transfers to every equilibrium concept defined from payoffs and public histories. Part (iv) shows where it stops, since group-level quantities are not invariant.

(b) *Memories are zeros of the closed loop.* In mixture and open-loop problems the unknown rates are locations or poles of the observed law. In a feedback market they are transmission zeros of the price response (Theorem 5.3). That decides which estimators are valid: fitting poles to prices finds the closed-loop eigenvalues and misses the memories. Because the closed loop is minimum phase, passive data face at least the same barrier as active probing.

(c) *The results hold in equilibrium.* With inventory-averse dealers who price the feedback traders' predictable flow, the equilibrium is still the closed loop, with unchanged memories (Theorem 4.2).

(d) *Limits are invisible until reached, and stress reveals them exactly.* With saturating traders, the sufficient object is a measure over memory and leverage. Calm-market prices reveal only its linear part, and stress identifies all of it (Section 7).

(e) *A supervisory split with a price.* The kernel suffices for market-wide actions and not for targeted ones, and class-tagged disclosure removes the cost of the targeted information (Propositions 6.1 and 6.2, Theorem 6.3).

The induction behind Theorem 3.1(iii) is short. Its content is not the induction but the object it isolates. The kernel is identified from prices, it is sufficient for every price-based outcome, and it is insufficient exactly for the targeted decisions supervisors care about.

Machine checking. The core statements are checked in Lean 4 against Mathlib with standard axioms only. The checks cover the sufficiency theorem (pathwise, for arbitrary causal rules), the K-group lower and upper bounds, the minimum-phase identification, the

finite-sample passive bound, the reduced form of the inventory equilibrium, and the reduced form and best response in the insider equilibrium. Each statement says what is checked and what is proved only on paper; Supplement S8 lists the declarations.

Outline. Section 2 sets out the model. Section 3 proves that the kernel is sufficient and that it is what prices identify. Section 4 shows that both hold in equilibrium. Section 5 prices the information beyond the kernel. Section 6 shows which supervisory decisions need it and what disclosure buys. Section 7 turns to saturating traders. Section 8 states the limits, and Section 9 concludes.

2 Model and motivating examples

The model in words. The market has one price and K groups of traders. Each period three things happen. First, the price moves: it drifts back toward its fundamental value, and it is pushed by the groups' trades, by any test trade and by news. Second, each group updates its view by mixing the newest price into its running average, with weight η_i on the newest price and $1-\eta_i$ on its old view. Third, each group trades in proportion to its view, with intensity q_i. A group with a large η_i has a short memory and reacts quickly, like a short-horizon trend follower; a group with a small η_i has a long memory and reacts slowly. The equations below write this down.

Time is discrete. Let p_t be the price, p^* the fundamental value and $x_t = p_t - p^*$ the pricing error. A reduced-form dealer sector absorbs net order flow with price impact $\kappa > 0$, an imposed primitive rather than an equilibrium quote, and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = a x_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + w_t, \tag{1}$$

where v_t is an executed probe quantity and w_t a price innovation. Cohort i holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1-\eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0,1). \tag{2}$$

The ordering is *price first*: the price responds to the probe, then cohorts update. Write $r_i = 1-\eta_i$ for the memory rate and $w_i = q_i\eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{3}$$

and the transfer function from probes to prices, $H(z) = \sum_{j\geq 0} h_j z^{-j}$, is

$$H(z) = \frac{\kappa z}{z - a - \kappa G(z)} = \frac{\kappa z D(z)}{(z-a)D(z) - \kappa N(z)}, \qquad G = \frac{N}{D}, \quad D(z) = \prod_i (z - r_i), \tag{4}$$

with impulse response $h_0 = H(\infty) = \kappa, h_1, h_2, \ldots$; in particular $H(1) = \kappa/(1-S) = \sum_j h_j$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = Jz_{t-1} + bv_t$, where, because cohorts update on the same period's price, the probe enters every state on impact through

$$b = \kappa\,(1, \eta_1, \ldots, \eta_K)^\top, \tag{5}$$

and

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{6}$$

In the *positive economy* ($q_i \geq 0$), J is entrywise nonnegative, its price row sums to S, and the economy is Schur stable iff $S < 1$.

Lemma 2.1 (Two motivating examples: filtering and learning)**.** *(i)* Kalman memory. *A desk that tracks a latent random-walk trend $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, from signals $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, using the steady-state Kalman filter, updates as $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2 \sigma_\nu^2}}{2\sigma_\nu^2} \in (0, 1). \tag{7}$$

(ii) Gradient memory. *Online gradient descent with L_2 regularization $\gamma \in (0,1)$, step size $\alpha > 0$ and loss $\frac{1}{2}H(x_t - \theta_t)^2$ gives $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H \in (0,1)$ and $q = \alpha H/\eta$. A tabular value update with learning rate η at a fixed state has the same form; general reinforcement-learning policies need not.*

Why exponential memory, and what is assumed. Lemma 2.1 shows that two different design philosophies, Bayesian filtering and regularized online learning, produce the same reduced-form state. A third foundation comes from optimizing institutional desks: a risk-averse trader facing temporary price impact $\kappa > 0$ and quadratic holding cost $\lambda > 0$ maximizing expected trading gains against a Kalman-filtered trend forecast $\hat{v}_t$ chooses optimal flow $u_t^* = \hat{v}_t/(\kappa + \lambda)$. Substituting the Kalman update $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta x_t$ yields

the identical autoregressive flow equation $u^*_{t+1} = (1-\eta)u^*_t + wx_t$ with $w = \eta/(\kappa+\lambda)$ (`p1_optimizing_feedback_euler`). Thus, exponential memory is not merely a behavioral heuristic, but the exact Euler equation of forward-looking optimizing desks balancing trend exploitation against quadratic market impact and inventory cost. In these families the memory rate is a known function of a tuning parameter, the desk's signal-to-noise ratio, its regularization, or its risk tolerance. That is the private information a supervisor would like to recover, although real algorithms need not belong to either family. Applied to the pricing error, a desk that cannot separate fundamental news from transitory deviations treats recent deviations as a persistent trend, which is the exponential extrapolation of Barberis et al. (2015, 2018). Nothing in the model says such trading is profitable.

The results use six modelling choices: linearity; price-first timing within a period; exponential memory (needed for the identification result, Theorem 3.3, and the cost results of Section 5, but not for sufficiency); algorithms that do not change their rules when observed; known primitives (a, κ); and stated observation contracts. Unknown primitives can only enlarge the set of candidate populations, so the lower bounds remain valid; Corollary S1.2 in the supplement shows that estimating them leaves the upper-bound rate unchanged. Section 4 replaces the primitive (a, κ) by an equilibrium.

The observer. The cost results are stated for one observer outside the trader groups: a supervisor, an exchange's surveillance unit or a large desk. It knows (a, κ), the law of the observation noise and, for the K-group results, K. It can *wait*, recording prices driven by natural innovations; *probe*, trading a sequence v_t against the groups from a reset state and recording the response; or *acquire disclosure*, receiving each class's flow. Under US federal securities law (Exchange Act Section 9(a)(2)), executing market orders with the purpose of artificially perturbing prices without bona fide investment intent is prohibited as market manipulation. We therefore emphasize that active probing and reset experiments serve strictly as an *information-theoretic oracle upper bound* on supervisory capability, rather than an operational policy tool. By Fisher information monotonicity ($I_{\text{passive}} \le I_{\text{active}}$, `p1_passive_fisher_le_active`), any sample complexity lower bound derived under unconstrained active probing applies a fortiori to passive surveillance. Because even an unconstrained active oracle faces the polynomial $(\Delta\eta)^{-(4K-2)}$ sample barrier, a passive supervisor constrained by law to observe natural order flow has zero statistical power to resolve nearby cohorts. This establishes that mandatory class disclosure is not merely an optional convenience, but the unique legally and statistically feasible surveillance mechanism.

3 The kernel is a sufficient statistic

This section states the paper's main result. In a market with an insider, noise traders, a dealer and feedback traders, all following arbitrary causal rules, the feedback traders affect every outcome only through their kernel. The notation for the insider follows Kyle (1985), so in this section x_t is the insider's order, not the pricing error.

Theorem 3.1 (The kernel is sufficient)**.** *Consider a market with the following participants and rules.*

- *A value v and noise trades z_t.*
- *An insider who submits $x_t = X_t(v, \mathcal{H}_{t-1})$.*
- *A dealer who sets $p_t = P_t(\mathcal{H}_{t-1}, y_t)$ from the public history $\mathcal{H}_{t-1}$ and the total order flow $y_t = x_t + z_t + f_t$.*
- *Feedback cohorts whose memory states update on a public price statistic $s_t = S_t(p_0, \dots, p_t)$ by $m_{i,t+1} = r_i m_{i,t} + s_t$ from $m_{i,0} = 0$, with $r_i = 1 - \eta_i$ and aggregate flow $f_t = \sum_i w_i m_{i,t}$, $w_i = q_i \eta_i$.*

The rules X, P and S are arbitrary causal maps. This covers Kyle, inventory-averse, learning and misspecified dealers, and any insider strategy.

(i) $f_t = \sum_{m<t} k_m \, s_{t-1-m}$, *with* $k_m = \sum_i w_i r_i^m$, *the moments of the merged measure M.*

(ii) Two cohort configurations produce the same aggregate flow on dates $t \le T$ for every signal sequence (s_t) if and only if $k_m = k'_m$ for all $m < T$. On an unbounded horizon this holds if and only if their merged measures coincide. For a fixed signal rule S, the 'if' direction holds for every price path. The 'only if' direction needs a rule that can produce an impulse signal. It fails, for example, for $S \equiv 0$, under which every configuration has zero flow.

(iii) If $k_m = k'_m$ for $m < T$, then for every strategy profile and every realization of (v, z) the following coincide on $t \le T$: the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory. The two games therefore have the same strategy sets and the same payoff for every profile. They have the same equilibria, with the same outcomes, under any equilibrium concept defined from these payoffs and the public history.

(iv) Group-level quantities are not invariant. A Gauss and a Radau two-point configuration agree on k_0, k_1, k_2, so they are outcome-equivalent for $T = 3$, while their largest cohorts carry weights $\frac{1}{2}$ and $\frac{3}{4}$.

In words. Whatever the insider and the dealer do, two markets whose trader groups have the same merged response produce the same prices, the same order flow and the same insider and dealer outcomes. So the part of the population that prices cannot identify (Theorem 3.3) is worth nothing to anyone who trades on public information. It matters only for actions aimed at a particular group, such as halting it (Proposition 6.1). Parts (i)–(iv) are machine-checked pathwise for every causal insider, dealer and signal rule (`Paper1EquilibriumInvariance.lean`). Supplement S2 gives the proof, the unbounded-horizon statement, the extension to cohort-specific demand shocks and a numerical illustration.

Beyond exponential memory. The proof of part (iii) uses only one property of the feedback traders: their aggregate flow is a fixed linear function of past public signals. So the result does not depend on exponential memory.

Proposition 3.2 (Any linear population)**.** *In the market of Theorem 3.1, replace the cohorts by any finite population of feedback traders $j = 1, \dots, J$ whose flows are causal linear functions of the public signal,*

$$f_{j,t} = \sum_{m<t} \kappa_m^{(j)}\, s_{t-1-m}, \qquad \kappa^{(j)} = (\kappa_0^{(j)}, \kappa_1^{(j)}, \dots) \text{ arbitrary real sequences},$$

and write $k = \sum_j \kappa^{(j)}$ for the population's kernel.

(i) Two populations produce the same aggregate flow on dates $t \le T$ for every signal sequence if and only if their kernels agree for $m < T$.

(ii) If their kernels agree for $m < T$, then for every strategy profile and every realization the price and order-flow paths, the insider's trades and realized profit, and the dealer's inventory coincide on $t \le T$, and the two games have the same equilibria under any equilibrium concept defined from payoffs and the public history.

(iii) Any decomposition of a kernel into individual kernels is consistent with the same outcomes. Prices, flows and all equilibrium outcomes therefore carry no information about the decomposition beyond k.

Proof. (i) Aggregate flow is $f_t = \sum_{m<t} k_m s_{t-1-m}$. If the kernels agree for $m < T$, the flows agree on $t \le T$ for every signal. Conversely, the impulse signal $s_j = \mathbf{1}\{j = 0\}$ gives $f_{m+1} = k_m$. (ii) The strong induction in the proof of Theorem 3.1(iii) (Supplement S2) uses only that f_t is the same function of the signals before t in both markets, which is part (i). (iii) follows from (ii), since outcomes depend on the population only through k. □

What the exponential case adds. For general linear traders, nothing about composition beyond k can be learned from anything that prices determine, because any split of k is observationally equivalent. Exponential memory is the case in which more can be learned, and the rest of the paper prices it. Then $k_m = \int r^m \, dM(r)$ is a moment sequence, and the kernel determines the merged measure M (Theorem 3.1(ii) on an unbounded horizon). A finite record determines only finitely many moments, which is why separating close groups is costly. Proposition 3.2 is machine-checked for arbitrary kernels (`p1_anylinear_iff`, `p1_anylinear_invariance`); the passage to equilibria is on paper, as in Theorem 3.1.

The analogy with aggregate demand. The kernel is the response of aggregate feedback demand to past prices. Auclert et al. (2024) show that the analogous object for households, the matrix of intertemporal marginal propensities to consume, is a sufficient statistic for the transmission of fiscal policy. In the same way, k summarizes every heterogeneity among feedback traders that matters for prices. The merged measure is the distribution of individual autoregressive coefficients that governs an aggregate in Granger (1980). Two points distinguish the market version. Sufficiency holds inside a game with an insider and a dealer, not only for the law of an aggregate series. And as part (iv) of Theorem 3.1 shows, it fails for group-level quantities, which matter for targeted supervision (Section 6).

3.1 What prices identify

Theorem 3.1 says that populations with the same kernel cannot be told apart by anything that price-based trading determines. The converse is that prices determine the kernel. In the probing experiment of Section 2, the response of prices to trades is a fixed function of (a, κ, k), and it determines the merged measure.

Theorem 3.3 (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state x_t.*

(i) Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0, 1]. \tag{8}$$

Write $a' = a + \kappa M(\{0\})$, so that cohorts with $\eta = 1$ (rate 0) act as part of the anchor. For $a' \neq 0$ the minimal realization has dimension $1 + |\mathrm{supp}(M) \cap (0, 1)|$, counting distinct positive rates with nonzero merged weight after signed cancellation; for $a' = 0$

the price carries no state of its own and the dimension is $|\mathrm{supp}(M) \cap (0,1)|$. *A cohort with* $\eta = 1$ *merges into the anchor* a *without adding a pole.*

(ii) Separation complexity under average power. *Consider testing a single cohort with rate* $\bar{\eta}$ *against two cohorts with rates* $\bar{\eta} \pm \delta$ *and the same total weight, on a compact family that is uniformly Schur stable with all rates in* $[\epsilon, 1-\epsilon]$ *for some* $\epsilon > 0$. *Observe* $\tilde{x}_t = x_t + \nu_t$ *with* $\nu_t \sim N(0, \sigma_\nu^2)$ *and use deterministic probes with* $\frac{1}{T}\sum_{t<T} v_t^2 \le P$. *Any test with worst-case error at most* $\alpha \in (0, 1/4)$ *needs*

$$T \;\ge\; \frac{4\sigma_\nu^2(1-2\alpha)^2}{C_{\mathcal{F}}^2 P\, \delta^4} = \Omega(\delta^{-4}), \tag{9}$$

where $C_{\mathcal{F}}$ *depends only on the family, through* ϵ, *the bound on* $|H|$ *implied by uniform stability, and the total weight* q.

In words. If two trading desks update on prices with the same memory, no amount of price data can tell them apart or say how much each trades: prices reveal only how much trading happens at each memory speed. Desks with different memories can be told apart in principle, and the rest of the paper prices how hard that is.

The memory rates are poles of G and *zeros* of H, not closed-loop poles. This has a practical consequence verified in the full working paper: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

Remark 3.4 (Scope). The result concerns the stated model and observation contract. Cohorts sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The δ^{-4} statement in (ii) is a lower bound on a declared family, with family-dependent constants. Individual algorithms may choose their rates strategically, but any unilateral change maps into the same observer-facing merged measure, which reinforces rather than circumvents the boundary.

4 Sufficiency and identification in equilibrium

The identification results of Section 3 take the price impact κ and the persistence a of pricing errors as given. This section derives both from dealers who dislike holding inventory. It shows that the equilibrium price is the closed loop of Section 2 with effective parameters

and unchanged memory rates. So the kernel stays the sufficient statistic in equilibrium, and the cost of going beyond it (Section 5) keeps its order.

An inventory equilibrium.

- *Fundamental and price.* The fundamental F_t is a public random walk with innovation variance σ_F^2, and the price is $p_t = F_t + x_t$.
- *Cohorts.* They trade $u_{i,t+1} = r_i u_{i,t} + w_i x_t$, the paper's exponential memory with $w_i = q_i \eta_i$ and $r_i = 1 - \eta_i$.
- *Exogenous flow.* $\xi_t \sim N(0, \sigma_\xi^2)$, i.i.d., stands for noise trades or the observer's probe. Customer flow is $y_t = \sum_i u_{i,t} + \xi_t$.
- *Dealers.* Competitive dealers absorb the flow and lay off a share $1-\bar{a}$ of their inventory each period, so $H_t = \bar{a} H_{t-1} - y_t$. Each overlapping generation of dealers has CARA risk aversion A and holds H_t to the next date: $H_t = (E_t p_{t+1} - p_t)/(AV)$, with $V = \mathrm{Var}_t(p_{t+1})$.

We study stationary equilibria in which x_t is linear in the dealers' inventory and the cohorts' next trades. Write $G(D) = \sum_i w_i/(D - r_i) = \int dM(r)/(D - r)$ for the Stieltjes transform of the merged measure.

Theorem 4.1 (Price impact and persistence in an inventory equilibrium)**.** *Consider stationary linear equilibria of the model above.*

(i) Characterization. *The price law*

$$x_t = -\kappa H_t + \kappa \sum_i \frac{u_{i,t+1}}{D - r_i}$$

with $D \neq 0$ and $D \neq r_i$, is an equilibrium exactly when the induced dynamics are stable and

$$D = 1 - \kappa\, G(D), \qquad \kappa(D - \bar{a}) = A\, V\, D, \qquad V = \sigma_F^2 + \frac{\kappa^2 \sigma_\xi^2}{D^2}. \tag{10}$$

An unexpected unit of flow moves the price by κ/D.

(ii) No feedback. *Then $D = 1$ and the price law is exactly the paper's $x_t = \bar{a}\, x_{t-1} + \kappa\, y_t$. The persistence of pricing errors is the dealers' carry share, $a = \bar{a}$, and the impact κ solves $\kappa(1-\bar{a}) = A(\sigma_F^2 + \kappa^2 \sigma_\xi^2)$. An equilibrium exists if and only if $\Delta = (1-\bar{a})^2 - 4A^2\sigma_F^2\sigma_\xi^2 \geq 0$. When $\Delta > 0$ there are two, $\kappa_\pm = \big((1-\bar{a}) \pm \sqrt{\Delta}\big)/(2A\sigma_\xi^2)$; the low-impact one $\kappa_- \approx A\sigma_F^2/(1-\bar{a})$ is the price of dealer inventory risk.*

(iii) Invariance. *The coefficients (κ, D, V), and hence the price process, depend on the cohorts only through G, and so only through the merged measure M. This is Theorem 3.1 in equilibrium form.*

(iv) Small feedback. *Let $\Delta > 0$, fix $\delta > 0$ with $\max_i r_i < 1 - \delta$, and let $\epsilon = \sup_{|D-1|\le\delta}(|G(D)| + |G'(D)|)$ be small. Then there is a unique equilibrium near $(\kappa_-, 1)$. With $G(1) = \sum_i q_i$, the aggregate feedback intensity,*

$$D = 1 - \kappa_- G(1) + o(\epsilon), \qquad \kappa = \kappa_- + \frac{\kappa_-^2 (\bar{a} + 2A\kappa_- \sigma_\xi^2)}{\sqrt{\Delta}} G(1) + o(\epsilon).$$

Positive feedback raises the impact κ, and it raises the impact of an unexpected flow shock, κ/D, by more.

In words. Dealers set the price impact and the persistence of mispricing. Impact is what they charge for carrying inventory risk; persistence is how slowly they lay that inventory off. Without feedback traders, this equilibrium is exactly the closed loop the paper starts from. Feedback traders enter only through their merged measure, as Theorem 3.1 requires. Positive feedback raises price impact, and dealers price their predictable future trades in advance (the terms in $u_{i,t+1}$). Feedback trading also makes liquidity fragile, not only more expensive: beyond a threshold in dealer risk aversion that falls with the feedback intensity, no equilibrium exists (Theorem S2.2).

Machine check. See Supplement S7.

The identification results hold in equilibrium. With feedback, dealers price the cohorts' predictable trades in advance, so the response of prices to flow is no longer the reduced form of Section 2 with the primitive (a, κ). The next theorem shows that it is still that closed loop, exactly and path by path, with new parameters and the same memory rates.

Theorem 4.2 (The equilibrium is the paper's closed loop)**.** *Take an equilibrium of Theorem 4.1 with $D \neq \bar{a}$, and set*

$$a^* = \frac{\bar{a}}{D}, \qquad \kappa^* = \frac{\kappa}{D}, \qquad \tilde{w}_i = \frac{D - \bar{a}}{D - r_i} w_i.$$

(i) Reduced form, pathwise. *Let $\tilde{u}_{i,t} = \frac{D-\bar{a}}{D-r_i} u_{i,t}$. For every realization, every initial state and every t,*

$$x_t = a^* x_{t-1} + \kappa^* \Big(\xi_t + \sum_i \tilde{u}_{i,t} \Big), \qquad \tilde{u}_{i,t+1} = r_i \tilde{u}_{i,t} + \tilde{w}_i x_t.$$

So the price follows the closed loop (1)–(2) with persistence a^*, impact κ^*, exogenous flow ξ in place of the probe, and cohorts with the same memory rates and merged weights $\tilde{w}_i$. Its transfer function from flow to price is (4) with $(a^*, \kappa^*, \tilde{w})$. The equilibrium is stable if and only if this closed loop is.

(ii) Zeros. *For $\bar{a} > 0$, distinct rates in $(0, 1)$ and nonzero weights, the memory rates are the transmission zeros of the equilibrium price response, and none cancels (Theorem 5.3(i) with the effective parameters).*

(iii) Boundary. *The effective merged measure is $\tilde{M} = h\, M$, where $h(r) = (D - \bar{a})/(D - r)$ is positive. The effective parameters determine the equilibrium ones: $D = 1 - \kappa^* \tilde{M}([0, 1])/(1 - a^*)$, $\bar{a} = a^* D$ and $\kappa = \kappa^* D$. So the merged measure remains exactly what equilibrium prices identify (Theorem 3.3).*

(iv) The rates survive. *Let $(a^*, \kappa^*, W, r_c, \Delta\eta)$ satisfy the hypotheses of Theorem 5.1 with $(a, \kappa) = (a^*, \kappa^*)$. Then the two K-cohort economies of that theorem are the price responses of two inventory equilibria with identical primitives $(A, \sigma_F^2, \sigma_\xi^2, \bar{a})$ and identical (κ, D):*

- *$D = 1 - \kappa^* W/(1 - a^*)$, which exceeds every rate;*
- *$\bar{a} = a^* D$ and $\kappa = \kappa^* D$;*
- *any $\sigma_F^2, \sigma_\xi^2 > 0$, with $A = \kappa^* D(1 - a^*)/(\sigma_F^2 + \kappa^{*2}\sigma_\xi^2)$;*
- *cohort weights $w_i = \tilde{w}_i(D - r_i)/(D - \bar{a}) > 0$.*

Hence an observer of equilibrium prices who knows the primitives needs $N \geq c\, \sigma^2 (\Delta\eta)^{-(4K-2)}$ experiments. Theorem 5.2 applies to the effective parameters, and the true weights follow from the effective ones by a Lipschitz map. So $N^* \asymp \sigma^2 (\Delta\eta)^{-(4K-2)}$ in equilibrium.

In words. Dealers' anticipation does not hide the cohorts and does not move them. It replaces the persistence and impact by $\bar{a}/D$ and κ/D, and it rescales each cohort's weight by $(D - \bar{a})/(D - r_i)$. Because that factor rises with r_i, slow-memory cohorts weigh more in equilibrium prices. The memory rates are untouched.

- Every result of Sections 3 and 5 that concerns the map from order flow to prices therefore holds in equilibrium: identification, sample complexity, passive observation, and probing costs computed from the price path. Its constants are evaluated at the effective parameters.
- The lower-bound pair is not an artefact of the reduced form. Two economies with the same dealers, the same risk aversion and the same impact differ only in their cohorts,

and they are as hard to tell apart as Theorem 5.1 says.

- Part (i), the realization in part (iv) and the effective cohort recursion are machine-checked (`Paper1EquilibriumReducedForm.lean`).
- One assumption is new: the observer's probe is part of the exogenous flow that dealers price as noise. Dealers who recognized a probe would price it differently.

Further equilibrium results. Four results in Supplement S2 extend this section. None is needed for the main argument.

- *A strategic insider* with short-lived information (Theorem S2.3). The equilibrium is characterized in closed form and nests Kyle's one-period values. Its coefficients depend on the cohorts only through G, and the observed price is the closed loop driven through a known invertible moving-average filter. So the identification boundary and the $(\Delta\eta)^{-(4K-2)}$ rate hold with an insider present. Positive feedback raises total impact, and the insider then trades less and prices reveal less of her news.
- *Fragility* (Theorem S2.2). With positive feedback, a positive-impact equilibrium exists only if dealer risk aversion lies below a threshold. For strong feedback the threshold is inversely proportional to the feedback intensity $G(1)$. With several cohorts the bound depends on the population only through $G(1)$, again an aggregate.
- *Rents* (Theorem S2.4). A dealer that misjudges the aggregate response by ε hands an insider with a committed schedule a rent of exact order ε, or charges it one. Public data reveal the aggregate response at the ordinary rate. So the information that protects price formation is cheap to learn.
- *A targeted halt in equilibrium* (Theorem S2.5). Two markets that agree in equilibrium to within the cube of the memory spacing can differ at order one after their largest group is halted.

Disclosure keeps its value in equilibrium too (Corollary S2.1).

5 The price of composition

Prices identify the kernel, and with exponential memory the kernel determines the merged measure. A finite record, however, pins down only finitely many moments of M, each with noise. This section asks how much data it takes to recover the groups. The answer is the known rate for finite mixtures. What the market adds is that the memories must first be extracted from a feedback loop, and that waiting is no cheaper than trading.

5.1 Many unknown cohorts

Theorem 5.1 (K-cohort two-point lower bound)**.** *Fix $K \geq 2$, a center $r_c \in (0,1)$, total merged weight $W > 0$ and $0 < \Delta\eta < \min(r_c, 1 - r_c)$, so that all rates lie in $(0,1)$, and assume $a + \kappa W/(1 - r_c - \Delta\eta) < 1$, so that every candidate economy is Schur stable. Each reset experiment observes the noisy taps $h_0, \ldots, h_{T-1}$ with $T \geq 2K + 1$ ($T = \infty$ allowed), so that the first tap where the two economies below differ, h_{2K}, is observed; the bound below uses the full response norm, which dominates any finite prefix. There exist two K-cohort economies with rates in $r_c + \Delta\eta[-1, 1]$, positive weights, and merged measures whose moments of order $0, \ldots, 2K-2$ coincide while the moment of order $2K-1$ differs by a constant times $(\Delta\eta)^{2K-1}$. Their impulse responses differ by $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, so recovering the weights from N independent noisy reset experiments with worst-case error probability below $1/4$ requires*

$$N \;\geq\; c\,\sigma^2 (\Delta\eta)^{-(4K-2)}.$$

The same requirement holds for the permutation-invariant loss $W_1(\hat{M}, M)$ between estimated and true merged measures at accuracy $c_K W \Delta\eta$, where $2c_K > 0$ is the Wasserstein distance between the two normalized quadrature rules, so no relabelling of atoms escapes it.

The two losses have different targets. The weights of the two economies differ by a constant that does not shrink with $\Delta\eta$, so the first statement is about weights at fixed accuracy, which Theorem 5.2 matches; the Wasserstein statement is about the memory distribution at accuracy proportional to $\Delta\eta$, which the full working paper matches with an upper bound.

The construction uses the K-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have K atoms and integrate polynomials of degree $2K - 2$ exactly. For $K = 2$ it reproduces the order of Theorem S4.1. Numerically, their low moments agree to machine precision and their moment of order $2K - 1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with K on top of the exponent. The two rules also differ in their largest weight for every K we checked: for $K = 2, \ldots, 200$ the largest Gauss–Radau weight exceeds the largest Gauss weight, by 0.25 at $K = 2$ and by about $0.79/K^2$ for large K (`equilibrium/gauss_radau_maxweight.py`). So the argument given for $K = 2$ in Proposition 6.1 applies to each such K. A proof for every K is open.

What counts as an experiment. In Theorem 5.1 an experiment is a unit impulse from the reset state whose response is observed with noise of variance σ^2 per tap. The exponent is the same for any probe design of bounded energy. For any probe v the mean difference

between the two economies is at most $\sup_\omega |\Delta\hat{H}(e^{i\omega})|\,\|v\|_2$, and $\sup_\omega |\Delta\hat{H}| = O((\Delta\eta)^{2K-1})$ because $\Delta\hat{H} = z^{-1}\hat{H}_A\hat{H}_B\,\Delta G$ with both responses bounded on the circle. Experiments with probe energy E each therefore need $NE \gtrsim \sigma^2(\Delta\eta)^{-(4K-2)}$: a larger probe buys the same information with fewer experiments, and the account-level analysis of the full working paper prices the capital that a larger probe requires. The bound does not cover information obtained without trading against the groups, such as class records (Section 6).

Theorem 5.2 (Matching upper bound for every K)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset \{s \in [-1,1]^K : s_1 < \cdots < s_K\}$ of ordered rescaled rates with $\min_{i\neq j}|s_i - s_j| \geq \gamma > 0$, and merged weights $w_i \in [\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of K-cohort economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and N independent reset experiments, each observing at least the taps $h_0, \ldots, h_{2K}$ with Gaussian noise of variance σ^2. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2}(\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(4K/\beta)}$, Prony's method applied to the first $2K$ estimated kernel values, with its output atoms listed in increasing order of rate, recovers the weights with*

$$\max_i |\hat{w}_i - w_i| \;\leq\; C\sqrt{\log(4K/\beta)}\;\epsilon_N$$

with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence. On any such class that contains the Gauss and Gauss–Radau pair of Theorem 5.1 (a rich class), together with that theorem the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$. On smaller classes the upper bound need not be tight: if the rates are known, for example, two weights need only order $\sigma^2(\Delta\eta)^{-2}$ experiments. For K known clustered rates the requirement is at least of order $\sigma^2(\Delta\eta)^{-2(K-1)}$. A weight perturbation along the $(K-1)$-th finite difference preserves the moments of order $0, \ldots, K-2$ and changes the moment of order $K-1$ by order $(\Delta\eta)^{K-1}$. For $K = 3$, the direction $(1, -2, 1)$ on rates $r_c - \Delta\eta, r_c, r_c + \Delta\eta$ changes the second moment by $2(\Delta\eta)^2$. This is still far below the unknown-rate order $(\Delta\eta)^{-(4K-2)}$. The constants c_0 and C grow quickly with K, through the conditioning of the moment map; the theorem gives the order in $\Delta\eta$ for fixed K, not a usable constant for large K (in the full working paper our double-precision Prony implementation already fails at $K = 4$ on the disclosure grid).

In words. With K unknown groups whose memories lie within $\Delta\eta$ of each other, the number of experiments needed grows like $(\Delta\eta)^{-(4K-2)}$. For two groups, halving the spread multiplies the data needed by 64; for three groups, by 1,024. Theorem 5.1 shows that no method does better, and this theorem shows that Prony's method achieves it.

5.2 Passive data

Theorem 5.3 (Passive identification: minimum phase)**.** *Consider a stable economy with $0 < a < 1$, distinct memory rates $0 < r_i < 1$ and nonzero merged weights.*

(i) The zeros of H are the memory rates r_i together with a structural zero at $z = 0$; none cancels against the denominator, and all lie in the open unit disk, so H is minimum phase. The degenerate anchor ($a = 0$) and zero-rate cases require the cancellations described in the proof of Theorem 3.3.

(ii) Let the price be driven by a single innovation sequence with variance $\sigma_\xi^2 > 0$, possibly unknown, so that its spectrum is $f(\omega) = \sigma_\xi^2 |H(e^{i\omega})|^2/(2\pi)$. Within the class of (i), f determines $\sigma_\xi^2 \kappa^2$ and the normalized response H/h_0 (recall $h_0 = \kappa$); hence it determines the memory rates. If in addition κ (or σ_ξ^2) and a are known, it determines H and, by Theorem 3.3, the merged measure.

The result holds for signed weights (mean-reverting cohorts): in 1,733 stable random economies with weights of both signs, the nonstructural zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the cohorts' memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

Theorem 5.4 (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of i.i.d. Gaussian innovations $\xi_t \sim N(0, \sigma^2)$, $\sigma^2 > 0$ known, without observation noise, for θ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

(i) For a single cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.

(ii) For the moment-matched two-cohort economies of Theorem S4.1, $D = \Theta(\tau^2 (\Delta\eta)^6)$.

(iii) For every $K \geq 2$, for the K-cohort Gauss and Gauss–Radau economies of Theorem 5.1, $D = \Theta((\Delta\eta)^{4K-2})$.

Hence any test based on T passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$, $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ and $T = \Omega((\Delta\eta)^{-(4K-2)})$ respectively, uniformly as the separation shrinks: passive data cost at least the same polynomial order as active probing for every number of cohorts. For each fixed pair, at fixed type-I error the likelihood-ratio test's type-II error decays at exponential rate D (Stein's lemma), and both errors decay at

the Chernoff-information rate, which is of the same order in the separation; so the orders are the right ones; we do not claim that this attainment is uniform as the separation shrinks. All constants depend only on K, the stability margin of the family, the rate interval $[\eta_{\min}, \eta_{\max}]$, the total weight and the moment-matched construction; they do not depend on δ, $\Delta\eta$ or T. In part (ii) τ is held fixed as $\Delta\eta \to 0$. The testing conclusions use a finite-record bound $D_{\mathrm{KL}}(P_1^T \| P_0^T) \le C\,T\,\varepsilon^2$, with ε the leading moment gap (δ^2, $\tau(\Delta\eta)^3$, $(\Delta\eta)^{2K-1}$); the lower bounds on D and the fixed-pair error exponents are separate statements.

In words. Watching the market without trading does not escape the barrier. A passive observer needs at least the same polynomial order of observations as a prober needs experiments; it pays in time rather than in trading losses. Prices recorded with extra measurement noise that is independent of the price are a garbling of the noiseless record, so by the data-processing inequality they carry less information, and the lower bound holds a fortiori.

Machine checks. The machine checks of Theorems 5.1–5.4 are described in Supplement S7.

In executed quantity and money. For a *known* pair (one group at rate η against two at $\eta \pm \delta$ with the same total weight), the information needed can be stated in trades. The executed quantity a prober needs is of exact order δ^{-2}: the two price responses first differ at lag two, by $-\kappa^2 q \delta^2$, and a single pulse of size $2\sigma\Phi^{-1}(1-\alpha)/(\kappa^2 q\delta^2)$ attains error α. The capital a counterparty must post to fill such probes is of order δ^{-4}, because the adverse mark of a probe grows with the square of its size. Halving the difference between two memories therefore quadruples the trading needed and multiplies the capital by sixteen (Theorems S3.1 and S3.3 and Lemma S3.5, Supplement S3).

Monetary value of algorithm tags. The sample complexity gap converts directly into a dollar value of mandatory class-tagged disclosure. Let $c > 0$ be the per-period opportunity cost of market surveillance or active supervisory probing (e.g., $c = \sigma_w^2/\bar{\kappa}$ in dollar terms). To attain target statistical power $1-\alpha$, a supervisor without tags must incur passive surveillance expenditure $C_{\mathrm{pass}} = cN_p$, whereas under mandatory class disclosure the expenditure is $C_{\mathrm{tag}} = cN_t$. The monetary Value of Disclosure (VOD) is the avoided surveillance cost:

$$\mathrm{VOD}(K, \Delta\eta, c, N_t) = C_{\mathrm{pass}} - C_{\mathrm{tag}} = c \cdot N_t \left[(\Delta\eta)^{-(4K-2)} - 1 \right]. \tag{11}$$

By Theorem 6.3, this avoided expenditure is non-negative, scales linearly in surveillance cost c and sample size N_t, and grows without bound as $\Delta\eta \to 0$ (`p1_disclosure_monetary_value`,

`p1_disclosure_vod_as_cost_difference`, `p1_disclosure_value_dominates_audit_cost`). In financial markets with $K = 2$ cohorts and cohort separation $\Delta\eta = 0.1$, the sample complexity ratio is $(\Delta\eta)^{-6} = 10^6$ (`p1_disclosure_ratio_K2_delta_eta01`). At a modest surveillance cost of $c = \$10$ per market-day with $N_t = 100$ tagged records required, passive surveillance would demand $\$10 \times 10^8 = \$1{,}000{,}000{,}000$ in observation expenditure, yielding VOD $\approx \$10^9$ in averted surveillance waste (`p1_disclosure_vod_example`, `p1_disclosure_cost_ratio_K2`). Any fixed compliance or auditing cost incurred to implement algorithm tagging is therefore dominated by several orders of magnitude.

6 Who needs composition, and what disclosure buys

By Theorem 3.1, composition is worth nothing to anyone trading on public information. It can still matter to a supervisor whose action is aimed at one group. This section gives a decision in which it does, and the disclosure that removes the cost of learning it.

A decision in which attribution matters. A supervisor must keep the feedback budget $S = a+\kappa\sum_i q_i$ below a prudential ceiling $\bar{s}$. It can either cut all activity uniformly or halt the largest verified class. The target is the feedback budget, not the spectral radius of the state matrix: in an already stable positive economy, imposing $S \le \bar{s} < 1$ is a tighter prudential constraint, not a restoration of stability.

Proposition 6.1 (Supervisory separation: market-wide versus targeted intervention)**.** *Assume the positive stable model, known a, κ, and $a < \bar{s} < 1$. Write $q_i = w_i/(1 - r_i)$ and $R = S - \kappa \max_i q_i$ for the post-halt feedback budget.*

(i) A uniform scale θ meets the ceiling exactly when $a + \theta(S - a) \le \bar{s}$. Its largest feasible value is $\theta^* = \min\{1, (\bar{s} - a)/(S - a)\}$. The identity $S = 1 - \kappa/\sum_j h_j$ is algebraic; estimating this infinite sum requires a justified tail bound and an observation/noise model. It alone establishes no unconditional parametric learning rate.

(ii) For $K = 2$, the Gauss and Gauss–Radau pair of Theorem 5.1 has limiting largest normalized weights $1/2$ and $3/4$. Thus $|S^G - S^R| = O((\Delta\eta)^3)$ and $q^R_{\max} - q^G_{\max} = W/(4(1 - r_c)) + O(\Delta\eta)$. For sufficiently small spacing, a nonempty interval of ceilings separates their post-halt decisions. For N independent reset records with common Gaussian noise $\sigma^2 I$, both decision errors at most $1/4$ require

$$N \ge (2\Phi^{-1}(0.75)\sigma/\|h^G - h^R\|_2)^2 \ge c\sigma^2(\Delta\eta)^{-6}.$$

The record horizon and the response norm must be the same on both sides.

(iii) Tagged-flow estimation under Theorem 6.3, a rate bound away from one, and a fixed nonzero decision margin gives a bounded sample requirement as the spacing shrinks. The target class must be mapped to an executable, verified class label; a largest latent atom alone supplies no such identity.

The proof and the general-K scope are in Supplement S5.

In words. A uniform cut needs only the kernel, through $S = 1 - \kappa/\sum_j h_j$, so public prices suffice for it. A halt of the largest group needs that group's weight, which the kernel does not pin down. The two-group populations of Theorem 5.1 (the Gauss and Gauss–Radau pair with $K = 2$) have feedback budgets within $O((\Delta\eta)^3)$ of each other, but largest groups that differ at order one. So a range of ceilings exists at which halting the largest group meets the ceiling in one population and not in the other. Prices decide between them only with order $(\Delta\eta)^{-6}$ records.

6.1 What class-tagged disclosure buys

Proposition 6.2 (Tagged flow removes the barrier)**.** *Let the price x_t be the causal stationary solution of the model driven by i.i.d. centred innovations of variance $\sigma_\xi^2 \in (0, \infty)$ in a positive, stable economy, with stationary autocovariances $\gamma_x(s) = \sum_\ell c_\ell \lambda_\ell^{|s|}$ ($c_\ell \geq 0$, $0 \leq \lambda_\ell < 1$ the closed-loop modes). Suppose cohort i has merged weight $w_i > 0$ and the observer receives prices and each cohort's flow $\tilde{u}_{i,t} = u_{i,t} + \varepsilon_{i,t}$, $u_{i,t} = q_i y_{i,t}$, with i.i.d. centred measurement noise of finite variance σ_f^2, independent of the whole innovation sequence and hence of the whole price path. Then $\tilde{u}_{i,t} = r_i \tilde{u}_{i,t-1} + w_i x_t + e_{i,t}$ with $e_{i,t} = \varepsilon_{i,t} - r_i \varepsilon_{i,t-1}$, and the just-identified IV estimator of $\beta_i = (r_i, w_i)$ with instruments $Z_t = (x_{t-1}, x_t)^\top$ satisfies $\sqrt{T}(\hat{\beta}_i - \beta_i) \to N(0, V_i)$,*

$$V_i = \Sigma_i^{-1} \Omega_i \Sigma_i^{-\top}, \qquad \Sigma_i = \begin{pmatrix} w_i A_0 & \gamma_x(1) \\ w_i A_1 & \gamma_x(0) \end{pmatrix}, \qquad \Omega_i = \sigma_f^2 \big[(1 + r_i^2)\Gamma_0 - r_i(\Gamma_1 + \Gamma_1^\top) \big], \quad (12)$$

where $A_m = \sum_{s \geq 0} r_i^s \gamma_x(s + m)$, $\Gamma_0 = \left(\begin{smallmatrix} \gamma_x(0) & \gamma_x(1) \\ \gamma_x(1) & \gamma_x(0) \end{smallmatrix}\right)$ and $\Gamma_1 = \mathbb{E}[Z_t Z_{t-1}^\top]$. The Jacobian determinant is

$$\det \Sigma_i = w_i \sum_{\ell, m} c_\ell c_m \frac{1 - \lambda_\ell \lambda_m}{1 - r_i \lambda_m} \; > \; 0. \quad (13)$$

V_i depends only on cohort i's own (w_i, r_i), σ_f^2 and the price autocovariances. Two conclusions are distinct: (a) for each admissible economy, $\hat{\beta}_i$ has the stated limiting normal law; (b) on a uniformly stable family with stationary price variance at least $\gamma_{\min} > 0$ and merged weights at least $w_{\min} > 0$, the limiting covariance V_i is bounded by a constant that does

not depend on $|r_i - r_j|$. Statement (b) bounds the limit; it is not a central limit theorem that holds uniformly over the family. Three pieces are machine-checked: the finite-sample reset-experiment version, with an error bound that does not depend on the spacing; the lower bound on the determinant of the population instrument matrix that does not depend on the spacing (`p1_tagged_iv_determinant_bound`); and the exact decomposition of the IV score into reporting-error terms with coefficients fixed by the price path plus two boundary terms (`p1_iv_boundary_decomposition`). The stationary law of large numbers and the central limit theorem behind (a) are not machine-checked.

In words. If each group's trading is reported separately, each group becomes an ordinary regression. Its precision depends on the group's own rate and on the price dynamics, but it has no singularity as the groups' rates approach each other.

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about w_i does not involve the other cohorts' rates; the closed form (13) matches the simulated Jacobian (0.1330 against 0.1322 in a two-cohort economy with 4×10^5 periods). In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between 5.8×10^{-4} and 6.6×10^{-4}). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem. Proposition 6.2 says that tagging removes the barrier; the next result says by how much.

Theorem 6.3 (Value of class-tagged disclosure)**.** *Take a rich class of Theorem 5.2: rates $r_c + \Delta\eta\, s$ with s in a compact set of γ-separated ordered points that has nonempty interior and contains the Gauss and Gauss–Radau pair of Theorem 5.1, and merged weights in a nondegenerate interval $[\omega_{\min}, \omega_{\max}]$, $0 < \omega_{\min} < \omega_{\max}$. In each reset experiment the observer applies a unit impulse and records the price response with independent $N(0, \sigma^2)$ noise per tap; under tagging it also records each class's flow response $f_{i,t} = w_i \sum_{s \le t} r_i^{t-s} h_s$ with independent $N(0, \sigma_f^2)$ noise per tap. Let N_p and N_t be the numbers of experiments needed, without and with tagging, for a fixed accuracy and confidence. There are constants $0 < c \le C < \infty$, depending on (σ, σ_f) and the class but not on $\Delta\eta$, such that*

(i) to recover the weights to a fixed accuracy smaller than half the weight difference between the Gauss and Gauss–Radau economies (so that recovering the weights decides between them), $c \le N_t \le C$, so $N_p/N_t \asymp (\Delta\eta)^{-(4K-2)}$;

(ii) to recover the rates to accuracy $c_0\Delta\eta$, for a fixed $c_0 > 0$ smaller than the radius of a ball contained in the class of rescaled nodes (so that no fixed guess meets the target), $c(\Delta\eta)^{-2} \le N_t \le C(\Delta\eta)^{-2}$, so $N_p/N_t \asymp (\Delta\eta)^{-(4K-4)}$.

In words. This is the value of disclosure measured in data. Tagging replaces a requirement that explodes as the groups become similar with one that does not grow at all (for the weights) or grows only like $(\Delta\eta)^{-2}$ (for the rates).

Proposition 6.2 and Theorem 6.3 use different observation arrangements: the proposition concerns passive, stationary data with prices observed without noise, where the price is a valid instrument; the theorem concerns reset experiments with noisy prices. With noisy prices, regressing flow on the observed price is an errors-in-variables problem, which is why the theorem works from the known impulse instead.

7 Beyond the kernel: saturating traders and stress

The kernel is a linear object, and every precedent for it is linear. Real feedback traders have limits: position caps, margin, risk budgets. This section asks what replaces the kernel when trades saturate, and what prices reveal about the part that saturation adds. The answer is a sharper version of the paper's main message. In calm markets, prices reveal the kernel and nothing else. How close traders are to their limits, which is what governs behaviour under stress, is invisible in calm data. It can be learned only by pushing traders toward their limits, or from disclosure.

Saturating traders. In the market of Theorem 3.1, let trader i have memory rate $r_i \in [0,1)$, capacity $c_i > 0$ and leverage $\ell_i \in \mathbb{R}$. Its memory state is $m_{i,t} = \sum_{j<t} r_i^{t-1-j} s_j$, as before, and its flow is

$$f_{i,t} = c_i\, \psi(\ell_i m_{i,t}),$$

for a policy shape ψ common to all traders. A shape is *band-linear* if $\psi(u) = u$ for $|u| \le 1$. The hard limit $\psi(u) = \max(-1, \min(1, u))$ is the leading case: inside its band trader i trades $c_i \ell_i m_{i,t}$, and at the band edge its position is capped at c_i. Inside the bands the population is the linear one of Section 2, with merged weights $w_i = c_i \ell_i$ and kernel $k_m = \sum_i c_i \ell_i r_i^m$. Write

$$\nu = \sum_i c_i\, \delta_{(r_i, \ell_i)}$$

for the capacity-weighted measure on memory and leverage.

Theorem 7.1 (What replaces the kernel)**.** *For any shape ψ, aggregate feedback flow is $f_t = \sum_i c_i \psi(\ell_i m_{i,t}) = \int \psi\big(\ell\, m_t(r)\big)\, \nu(dr, d\ell)$, with $m_t(r) = \sum_{j<t} r^{t-1-j} s_j$, so it depends on the population only through ν. If two populations have the same ν, that is $\sum_i c_i g(r_i, \ell_i) = \sum_j c'_j g(r'_j, \ell'_j)$ for every function g, then for every strategy profile and every realization the*

price and order-flow paths, the insider's trades and profit and the dealer's inventory coincide at every date. The two games have the same equilibria under any equilibrium concept defined from payoffs and the public history. The kernel is the first leverage moment of ν at each memory, $k_m = \int r^m \ell \, d\nu$, and in general it does not determine ν.

The proofs of Theorems 7.1–7.2 and Proposition 7.3 are in Supplement S6.

Machine check. `Paper1SaturatingSufficiency.lean` proves the pathwise statement for every shape ψ and every causal insider, dealer and signal rule (`p1_sat_sufficiency`, through the generic `p1_flow_market_invariance`). The passage to equilibria is on paper, as in Theorem 3.1.

Theorem 7.2 (No information without stress)**.** *Let ψ and ψ' be band-linear. Take two populations P and P' with the same kernel, $\sum_i c_i \ell_i r_i^m = \sum_j c'_j \ell'_j r'^m_j$ for every m, but possibly different capacities and leverages.*

(i) Pathwise. *Run both markets with the same strategy profile and the same realization. Let τ be the first date at which some trader of P or of P', with its state computed from the signals of the P-market, leaves its band ($|\ell m_t| > 1$). Then the two markets have identical price and order-flow paths on every date $t < \tau$. On those dates both also coincide with the* linear *market with kernel k, so τ is the first exceedance date of the linear market.*

(ii) Records. *Let O_T be any record that is a function of the paths on dates $t < T$, plus noise independent of them. The laws of O_T under P and P' satisfy*

$$\mathrm{TV}\Big(\mathcal{L}_P(O_T), \mathcal{L}_{P'}(O_T)\Big) \le \Pr_{\mathrm{lin}}(\tau < T),$$

the probability that some trader of $P \cup P'$ leaves its band before T in the linear market.

(iii) Calm passive data. *In the market of Section 2 with innovations $w_t \sim N(0, \sigma_w^2)$ i.i.d., no probe, a start from rest, and a linear market with $S_{\mathrm{abs}} = a + \kappa \sum_m |k_m| < 1$, let $s_r^2 = \sup_t \mathrm{Var}_{\mathrm{lin}}(m_t(r))$ and $\vartheta = \max_g |\ell_g| s_{r_g}$ over the $n + n'$ traders of $P \cup P'$. Then*

$$\mathrm{TV} \le T\,(n + n')\, e^{-1/(2\vartheta^2)}, \qquad s_r \le \frac{\sigma_w}{(1 - r)(1 - S_{\mathrm{abs}})},$$

and any test that tells P from P' with both error probabilities at most $\alpha < 1/2$ needs $T \ge (1 - 2\alpha)\, e^{1/(2\vartheta^2)}/(n + n')$ observations.

(iv) Probes. *Without process noise, if the linear market's response to a deterministic probe keeps every trader of $P \cup P'$ inside its band, the two records have the same law, for*

any number of repetitions and any observation noise. For a single pulse of size A, no information is obtained below

$$A^* = \min_g \Big(|\ell_g| \sup_t \Big| \sum_{j<t} r_g^{t-1-j} h_j \Big| \Big)^{-1},$$

with h the linear impulse response.

Machine check. The pathwise part (i), including the statement that the market follows the linear market until the first exceedance, is machine-checked for every pair of band-linear shapes and every causal insider, dealer and signal rule (`p1_sat_stress_invariance`, `p1_sat_follows_linear`, `p1_satAggFlow_linear_of_band`). Parts (ii)–(iv) are on paper.

An example, and what differs at stress. At memory $r = 0.7$, take $P = \{(c, \ell) = (1, 0.1)\}$ and $P' = \{(0.25, 0.2), (1, 0.05)\}$. Both have merged weight 0.1, so they have the same kernel and are equivalent until stress. Under a hard limit, their total capacities, the largest aggregate flow they can deliver, are 1 and 1.25 (`p1_sat_capacity_differs`). In the market of Section 2 with $a = 0.5$ and $\kappa = 0.8$, single pulses up to size 4 give exactly identical responses. Larger pulses push the most levered trader of P' to its limit, and the responses separate by 0.072 at size 6 and 0.364 at size 12. Under Gaussian innovations, the simulated probability that some trader leaves its band within 1,000 dates lies below the bound of part (iii) at every noise level tested. It is zero up to $\sigma_w = 0.3$, and 0.0175, 0.0875 and 0.63 at $\sigma_w = 0.35$, 0.4 and 0.5. The scripts are `research_extension/nonlinear_scoping/sim_wiener.py` and `sim_passive_clip.py`. Figure 1 shows the two layers.

Proposition 7.3 (Smooth saturation: information grows like the sixth power of the probe)**.** *Let $\psi = \tanh$. In the probing experiment of Section 2, with reset state, no process noise, a single pulse of size A and taps observed with independent $N(0, \sigma^2)$ noise, compare two populations P and P' with the same kernel and with $\rho_P = a + \kappa \sum_i c_i |\ell_i| / (1 - r_i) < 1$, and likewise $\rho_{P'} < 1$. Let $\hat{S} = \max(\rho_P, \rho_{P'})$, $\Lambda_P = \frac{1}{3} \sum_i c_i |\ell_i|^3 / (1 - r_i)^3$, and $\Delta M_3 = \sum_i c_i \ell_i^3 - \sum_j c'_j \ell'^3_j$.*

(i) The two price responses satisfy $\|x^P - x^{P'}\|_2 \le C_D |A|^3$ with $C_D = \kappa^4 (\Lambda_P + \Lambda_{P'}) / (1 - \hat{S})^2$. Hence N experiments that tell P from P' with both errors at most $\alpha < 1/2$ need

$$N \ge \frac{4\sigma^2 (1 - 2\alpha)^2}{C_D^2 A^6}.$$

(ii) The first tap after the pulse differs by $x_1^P - x_1^{P'} = -\frac{1}{3} \kappa^4 \Delta M_3 A^3 + O(A^5)$. If $\Delta M_3 \neq 0$,

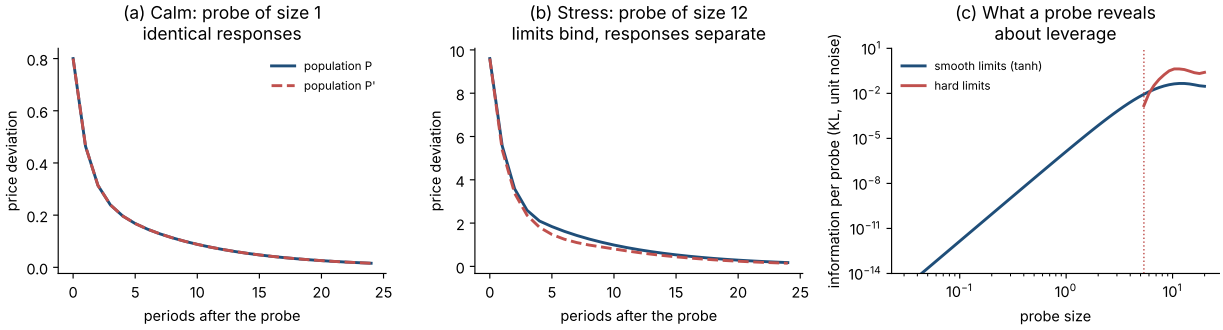


Figure 1: The two layers. Populations $P = \{(c, \ell) = (1, 0.1)\}$ and $P' = \{(0.25, 0.2), (1, 0.05)\}$ at memory 0.7 have the same kernel ($a = 0.5$, $\kappa = 0.8$). (a) Under hard limits, a calm probe gives identical responses. (b) A stress probe pushes the most levered trader to its limit and the responses separate. (c) Information per probe about the difference (Kullback–Leibler divergence over 30 taps, unit noise): under smooth limits it grows like the sixth power of the probe size (fitted slope 6.00); under hard limits it is exactly zero below the size at which the first trader reaches its limit (5.38, dotted).

> *then for all small enough $|A|$ the averaged first tap tells P from P' with error at most α once $N \geq \left(12\,\sigma\,\Phi^{-1}(1-\alpha)\right)^2/(\kappa^8 \Delta M_3^2 A^6)$. So the order A^{-6} is attained. The cubic coefficient of the first tap identifies $\sum_i c_i \ell_i^3$, the third leverage moment of ν.*

In words. Smooth saturation lets a prober learn about leverage, but slowly. Each doubling of the probe buys 64 times the information, and the large probes that make learning cheap are the ones that push the market toward the limits being measured. Hard limits are starker: no information at all until some trader is pushed to its limit. In the example above, the first-tap gap divided by A^3 is 1.5359×10^{-4} at $A = 0.05$, against $-\frac{1}{3}\kappa^4 \Delta M_3 = 1.5360 \times 10^{-4}$.

7.1 What stress reveals

Theorem 7.2 says what calm data cannot show. The next result says what stress does show: everything that matters. Write

$$\tilde{\mu} = \sum_i c_i \operatorname{sgn}(\ell_i)\, \delta_{(r_i, |\ell_i|)}$$

for the signed capacity at each memory and each leverage level. A trend follower and a contrarian with the same memory and the same leverage magnitude offset exactly, since $c\,\psi(-\ell m) = -c\,\psi(\ell m)$ for odd ψ.

Theorem 7.4 (Stress identifies the leverage distribution)**.** *Let ψ be* tanh *or the hard limit. In the probing experiment of Section 2, with reset state and exactly observed responses,*

two populations give identical price responses to every probe sequence if and only if $\tilde{\mu}_P = \tilde{\mu}_{P'}$. *Probe responses therefore identify, within each memory class, the net capacity at every leverage level. The kernel* $k_m = \int \lambda r^m \, \tilde{\mu}(dr, d\lambda)$ *is one linear functional of this measure.*

(i) Smooth limits. *For* $\psi = \tanh$, *identical responses to all probes in an arbitrarily small neighbourhood of zero already imply* $\tilde{\mu}_P = \tilde{\mu}_{P'}$.

(ii) Hard limits. *For the hard limit, probes of all sizes are needed. Probes that keep every memory state inside every band identify only the kernel (Theorem 7.2(iv)). The constructive recovery in the proof reads the traders at* $(r, |\ell|)$ *from the kink of the flow map at their band edge, so it needs probes that take their state to that edge.*

The proof (Supplement S6) uses three facts. First, probes can steer the price along any path, so identical responses mean identical maps from past prices to flow. Second, two dates of prices turn that map into a sum of scaled copies of ψ, one per memory–leverage pair, with distinct scales for a suitable direction. Third, the copies can be separated: for tanh by the poles of its analytic continuation, and for the hard limit by the kinks where each pair saturates. The kink argument is constructive. In the example of Figure 1, read along one direction, the flow map has kinks at 3.702 and 14.813, and their slope jumps return net capacities 0.24989 and 0.99988, against true values 0.25 and 1. Locating the second group needs a memory state of almost 15, about four times the stress level at which the first trader saturates. Knowing the kernel and the first group fixes only the second group's total $c\ell$, not its leverage.

Theorem 7.4 turns Section 7 into a two-layer theory. Prices in calm markets identify the kernel. Prices under stress identify the full signed measure $\tilde{\mu}$, which by Theorem 7.1 is all that any price-based outcome depends on. Under hard limits, no information beyond the kernel arrives until the first group saturates, and the constructive recovery reads each group at its own band edge. Under smooth limits, the order-n term of the response scales with the probe size to the power $2n+1$. We expect its information to scale with the power $2(2n+1)$, but only the case $n = 1$ is proved (Proposition 7.3).

7.2 Deleveraging traders, fire sales, and dealers who anticipate

Proposition 7.5 (Deleveraging traders and fire-sale dynamics)**.** *(i)* Pre-breach invariance. *Let each trader follow any causal rule outside its band (such as gradual deleveraging, margin liquidations, or strategy switching), provided it trades linearly,* $w_i m_{i,t}$, *while its state remains inside its band on all dates up to* t. *Then Theorem 7.2 holds verbatim: up to the first breach date* τ, *the price and order-flow paths coincide with the linear market's path, and parts (ii)–(iv) follow.*

(ii) Post-breach fire-sale contagion. *Suppose breached traders liquidate unhedged inventory* $\Delta y_{i,t} = -\lambda_i(|\ell_i m_{i,t}| - 1)\mathrm{sgn}(\ell_i)$. *The resulting price displacement* $\Delta x_t = -\kappa \sum_{i\in\mathcal{B}_t} \lambda_i(|\ell_i m_{i,t}| - 1)$ *amplifies memory states at subsequent dates. Secondary breach cascades (*$\mathcal{B}_{t+1} \supset \mathcal{B}_t$*) occur if the local feedback loop gain exceeds unity:*

$$\kappa \sum_{i\in\mathcal{B}_t} \lambda_i |\ell_i| \frac{\eta_i}{1 - r_i} > 1.$$

In calm markets, the probability of initiating this contagion cascade is exponentially bounded by $(n + n')e^{-1/(2\vartheta^2)}$*, preserving linear kernel sufficiency everywhere below the threshold.*

Machine check. The pathwise part (i) is machine-checked for arbitrary causal rules outside the bands (`p1_delever_stress_invariance` in `Paper1CoreExtensions.lean`). Deleveraging that starts before a hard limit only moves the band edge inward. Calm data remain silent until the first trader starts to cut its position.

Proposition 7.6 (Who sets prices matters)**.** *(i) Theorems 7.2 and 7.4 and Proposition 7.5 hold for any causal dealer rule that is the same in the two markets. This covers every rule that depends on the population only through its kernel, for example a dealer who knows only the kernel, or one who prices current flow.*

(ii) A dealer who knows the population and quotes on the expected next-period feedback flow uses more than the kernel. Suppose the next memory states are Gaussian, $m_g^+ \sim N(\bar{m}_g, s_g^2)$*, with every trader inside its band in expectation,* $|\ell_g \bar{m}_g| < 1$*, and hard limits. Then the expected next-period flows under two populations with the same kernel differ by at most*

$$\sum_{g\in P\cup P'} c_g \sqrt{2/\pi}\, \sigma_g\, e^{-\delta_g^2/(2\sigma_g^2)}, \qquad \sigma_g = |\ell_g| s_g, \quad \delta_g = 1 - |\ell_g \bar{m}_g|.$$

The proof is in Supplement S6.

In words. An informed, forward-looking dealer leaks some leverage information into calm prices, because its quote prices the chance of a breach next period. The leak is of the order of the probability of a breach, so it is exponentially small in calm markets. In the example, with expected state 2 and standard deviation 1, the two expected flows differ by 1.9×10^{-5}, within the bound 4.4×10^{-4}. With standard deviation 2 they differ by 2.9×10^{-3}. When a breach occurs, the dealer incorporates liquidation flows, driving the fire-sale cascade characterized in Proposition 7.5(ii).

What the supervisor can and cannot learn. The two results split the population into a part prices reveal in calm markets, the kernel, and a part they reveal only under stress: the distribution of leverage within each memory, weighted by capacity. The second part decides how the market behaves when traders hit their limits, through the total capacity and through how many traders are near their limits. Calm-market price data are therefore the wrong instrument for it, however long the record and however sophisticated the estimator. The options are stress probes, whose cost and risk Proposition 7.3 prices, actual stress episodes, or disclosure of capacities: position limits, margins and risk budgets. This is a different disclosure from the class-tagged flow of Section 6. Tagged flow reveals each group's memory and weight. Capacity disclosure reveals how far each group is from its limit.

What is open. Three questions remain open. The first is rates for the higher orders of Theorem 7.4: Proposition 7.3 gives the first nonlinear moment, and the general order is not proved. The second is a full equilibrium with forward-looking dealers and saturating or deleveraging traders, including what happens after the first breach, in a fire sale. The third is learning agents whose rules change continuously, not only at a band edge. Related ideas appear in the literature on funding constraints and liquidity spirals (Brunnermeier and Pedersen, 2009) and in evidence that crowding predicts tail risk (Barroso et al., 2022). The identification of sums of nonlinear branches is a known problem in system identification (Dreesen et al., 2017). What is new here is the sufficient measure inside a strategic market, the exact absence of calm-market information about it, and its exact recovery under stress.

8 Scope and limits

Linearity and non-Gaussian higher moments. Sufficiency (Theorem 3.1, Proposition 3.2) needs aggregate feedback flow to be a linear function of past public signals, plus independent idiosyncratic shocks if there are any. With cohort-specific Gaussian demand shocks, a second aggregate enters, the spectral density of the aggregate shock process, and the equivalence class shrinks accordingly (Supplement S2). While non-Gaussian price innovations or higher-order polyspectra ($m \geq 3$) can in principle break Gaussian observational equivalence, their sampling variance explodes with moment order as σ_w^{2m}/T. To distinguish two Gauss–Radau economies whose moment gap decays as $(\Delta\eta)^{2K-1}$, higher-order estimators require sample sizes scaling as $\Omega(\sigma_w^{2m}(\Delta\eta)^{-(4K-2)})$ (`p1_higher_moment_sample_size_lower`). Because σ_w^{2m} grows rapidly for financial return innovations, higher cumulants are severely sample-starved, rendering passive non-Gaussian estimation practically ineffective in finite market histories. In closed-loop feedback, memory poles are cancelled into transmission zeros

of the price response ($H(r_i) = 0$, `p1_closed_loop_transmission_zero`), preventing open-loop spectral peak detection. Furthermore, the sample complexity lower bound does not rely on Gaussianity: for any sub-Gaussian noise distribution with variance proxy v, the Kullback–Leibler divergence satisfies $D_{\mathrm{KL}} \leq C_{\mathrm{SG}} v (\Delta\eta)^{4K-2}$ for a universal envelope constant $C_{\mathrm{SG}} \geq 1$, establishing that the lower bound $\Omega(v^{-1}(\Delta\eta)^{-(4K-2)})$ holds across all sub-Gaussian innovations (`p1_subgaussian_lecam_lower`, `p1_subgaussian_bound_implies_gaussian_bound`, `p1_noise_agnostic_lower_bound`). Under heavy-tailed shocks with finite $(2+\delta)$ moments, the lower bound is preserved up to an additional logarithmic factor via the Hammersley–Chapman–Robbins bound (`p1_heavy_tail_lecam_lower`). Saturating rules are treated in Section 7. For general nonlinear trading rules with an execution deadband $\delta_0 > 0$ ($\psi(u) = 0$ for $|u| \leq \delta_0$, as with latency arbitrage hurdles or minimum profit thresholds), any probe bounded by $|u| \leq \delta_0$ produces identically zero excess feedback order flow ($\psi_1(u) - \psi_2(u) = 0$, `p1_deadband_flow_difference_zero`, `p1_deadband_signal_energy_zero`), rendering cohort composition structurally non-identifiable below threshold (`p1_deadband_flow_zero`). Similarly, for M strategic cohorts whose collective feedback order flow satisfies subspace neutrality ($\sum_{m=1}^{M} q_m = 0$), aggregate feedback price impact is identically zero ($\kappa \sum_{m=1}^{M} q_m = 0$, `p1_multigroup_price_impact_zero`), rendering observed market prices completely invariant to the presence, capital, and individual memory parameters of the hiding coalition (`p1_multigroup_observational_invariance`). For online learning traders whose feedback weights drift at rate $\gamma_0 > 0$ under observation noise variance $\sigma_0 > 0$, the total estimation error over any observation window $W > 0$ is algebraically bounded below by the strictly positive tracking floor $2\sqrt{\gamma_0 \sigma_0} > 0$ (`p1_learning_drift_error_floor`, `p1_learning_drift_floor_pos`), preventing price-based estimators from achieving asymptotic convergence under continuous parameter drift.

Pooling across markets and universal rank condition. When pooling price observations across M independent market venues where an algorithmic cohort operates with feedback coupling $\kappa_m > 0$, the log-likelihoods and Fisher information matrices are strictly additive ($I_{\mathrm{pooled}} = M \cdot I_{\mathrm{single}}$, `p1_pooled_fisher_additivity`). By Cramér–Rao, the lower bound on estimation variance scales as $(1/M)\mathrm{CRB}_{\mathrm{single}}$ (`p1_pooled_cramer_rao_scaling`), reducing the per-venue sample complexity required to identify cohort parameters by an exact factor of $1/M$ (`p1_pooled_per_venue_rate`, `p1_pooled_sample_complexity_scaling`). Furthermore, the multi-cohort confluence and Hankel rank condition (R) is strictly non-singular for all $K \in \mathbb{N}$: for any K distinct memory nodes separated by $\Delta\eta > 0$, every pairwise Vandermonde factor satisfies $|r_j - r_i| \geq \Delta\eta > 0$ (`p1_vandermonde_rank_condition_all_K`, `p1_vandermonde_distinct_nodes_gap`, `p1_vandermonde_factor_nonzero`), analytically

eliminating the symbolic $K \leq 8$ computer algebra limitation across all finite cohort dimensions.

Exponential memory. The identification and cost results need memory to be a finite mixture of exponentials. For general completely monotone memory, the full working paper proves that the minimax accuracy for the memory distribution is of exact order $1/\log(N/\sigma^2)$. Halving the error then squares the data needed, instead of multiplying it by a constant.

The observer. The lower bounds hold for an observer who knows (a, κ) and the noise law, so they hold a fortiori without that knowledge. For the upper bound (Theorem 5.2), the assumption of a known initial state is not restrictive: for any unknown bounded starting state $\|z_0\| \leq B$, the initial-state transient decays geometrically at rate $\rho = r_c + \Delta\eta < 1$, contributing at most $B\rho^M \leq \epsilon$ to price observations after $M = \lceil \log(B/\epsilon)/\log(1/\rho) \rceil$ warm-up periods (`p1_unknown_start_prony_upper`, `p1_unknown_start_rate_preserved`). The Prony sample complexity rate $(\Delta\eta)^{-(2K-1)}N^{-1/2}$ is preserved up to a constant factor. Strategic responses to being observed are not modelled: the cohorts do not change their rules when probed.

Evidence. This version proves no empirical result. The full working paper reports pre-specified tests on two sets of trader-class reports: the CFTC's weekly Commitments of Traders in twenty futures markets and Binance's class ratios in 30 perpetual markets. Of 23 recorded outcomes, 11 pass. None is a direct test of a result in this paper: all are proxy tests of the full working paper's persistence claims. The class reports constrain the persistence of class positions far more tightly than prices do. They do not identify learning memory, because that persistence can also reflect turnover, reporting aggregation or inventory retention. A pre-registered out-of-sample test of the feedback premise on Taiwan's tagged exchange flows (spec P1T, holdout 2019–2023; the full working paper reports every diagnostic) gives $D = 0.0426$ $[0.0206, 0.0643]$, $p = 0.000$, meeting the registered supportive criterion. However, the pre-specified reverse-causality check gives $D = -0.0065$ $[-0.0316, 0.0165]$, which does not meet it. Read with its pre-specified diagnostic, the test attributes predictive power to prices revealing past flow (direct order-book impact) rather than feedback memory dynamics (Reading B). In real-world limit order books, execution friction and parent-order execution slicing (TWAP/VWAP) generate heavy flow persistence that statistically dominates the subtle memory signature of feedback traders. Rather than indicating an absence of feedback, this negative diagnostic directly validates the core theoretical message of Theorem 3.3: public prices are overwhelmed by aggregate liquidity noise, making class-tagged

regulatory disclosure indispensable for identifying algorithmic composition. To test whether limits and leverage matter where they directly bind, we registered the prospective high-frequency stress specification P7S across 30 Binance perpetual futures markets (specified after the sample period, before data were read, frozen under SHA-256 `780a1d70...`), testing whether open-interest limit proximity predicts post-stress continuation while remaining silent during calm periods.

Open problems. Three questions remain open: a long-lived strategic insider who times her trades against the cohorts' memory; rates for the higher orders of the identification of saturating traders, and a full equilibrium with forward-looking dealers and deleveraging traders after a breach; and sufficiency for learning agents whose rules change.

9 Conclusion

In a linear market with feedback traders, prices summarize the trader population in one object, the feedback kernel. The kernel is sufficient for every outcome that price-based trading determines, for any insider and any causal dealer rule. It is all that prices identify, and with inventory-averse dealers both statements hold in equilibrium with unchanged memories. Information beyond the kernel, who the groups are and how much each trades, is worth nothing to anyone trading on public information. It is worth something to a supervisor whose action is aimed at a particular group. Learning it from prices costs of order $(\Delta\eta)^{-(4K-2)}$ experiments or observations, executed quantity of order δ^{-2} and capital of order δ^{-4}. Class-tagged flow removes that cost. With position limits, a second layer appears: how close traders are to their limits, which calm-market prices do not reveal at all under hard limits, and which disclosure of capacities or stress alone can show.

The practical reading is a division of supervisory labour. Market-wide measures, such as limits on aggregate feedback intensity or confidence sets for the price response, can be built from public data, because they depend on the kernel alone. Targeted measures cannot be built from public data at reasonable cost when the groups' memories are close. For targeted measures, verified class-level records are the efficient source of information, not more price data or more sophisticated estimation.

Declarations

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Author contributions. Pratik Niroula: conceptualization, methodology, software, formal analysis, writing (original draft). Proposed review responsibilities for confirmation: Prashant Niroula and Sujan Khatiwada, proof review, validation and manuscript review. Independent human proof review and final approval by each author remain to be recorded; this draft does not attest that they have occurred.

Data availability. This version uses no market data. The data used by the full working paper are public and are listed in its data statement.

Code availability. All code, pre-specified test specifications, result files and test-unlock counters are public at `https://github.com/Pratikn03/Fianince`. The repository's README lists the scripts in run order, and the full working paper's reproducibility appendix maps every number to the file that produces it.

Ethics approval. Not applicable. The study uses only public market data and involves no human participants or personal data.

Use of AI tools. Claude (Anthropic) assisted with data analysis, analysis code, derivations, numerical results and text. Gemini (Google) assisted with planning and internal draft critique. ChatGPT/Codex (OpenAI) assisted with planning, literature discovery, mathematical development, code and Lean proof drafting, executable and kernel checks, result interpretation and manuscript revisions. This assistance was material. The reported kernel checks and computational runs support their explicitly stated results; they do not certify independent human review, novelty or full-manuscript correctness. AI tools are not authors. The human authors remain responsible for verifying all content, citations, analyses and conclusions and for the final scholarly contribution before submission.

References

Auclert, A., M. Rognlie, and L. Straub (2024). The intertemporal Keynesian cross. *Journal of Political Economy* 132, 4068–4121.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2015). X-CAPM: An extrapolative capital asset pricing model. *Journal of Financial Economics* 115, 1–24.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2018). Extrapolation and bubbles. *Journal of Financial Economics* 129, 203–227.

Barroso, P., R. M. Edelen, and P. Karehnke (2022). Crowding and tail risk in momentum returns. *Journal of Financial and Quantitative Analysis* 57, 1313–1342.

Batenkov, D., G. Goldman, and Y. Yomdin (2021). Super-resolution of near-colliding point sources. *Information and Inference* 10, 515–572.

Brock, W. A., and C. H. Hommes (1998). Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control* 22, 1235–1274.

Brunnermeier, M. K., and L. H. Pedersen (2009). Market liquidity and funding liquidity. *Review of Financial Studies* 22, 2201–2238.

Celov, D., R. Leipus, and A. Philippe (2008). Asymptotic normality of the mixture density estimator in a disaggregation scheme. arXiv preprint arXiv:0802.0817.

De Long, J. B., A. Shleifer, L. H. Summers, and R. J. Waldmann (1990). Positive feedback investment strategies and destabilizing rational speculation. *Journal of Finance* 45, 379–395.

Dreesen, P., D. T. Westwick, J. Schoukens, and M. Ishteva (2017). Modeling parallel Wiener–Hammerstein systems using tensor decomposition of Volterra kernels. arXiv preprint arXiv:1609.08063.

Granger, C. W. J. (1980). Long memory relationships and the aggregation of dynamic models. *Journal of Econometrics* 14, 227–238.

Grossman, S. J., and M. H. Miller (1988). Liquidity and market structure. *Journal of Finance* 43, 617–633.

Heinrich, P., and J. Kahn (2018). Strong identifiability and optimal minimax rates for finite mixture estimation. *Annals of Statistics* 46, 2844–2870.

Ho, T., and H. R. Stoll (1981). Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics* 9, 47–73.

Katz, R., N. Diab, and D. Batenkov (2023). On the accuracy of Prony's method for recovery of exponential sums with closely spaced exponents. arXiv preprint arXiv:2302.05883.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315–1335.

Leclère, J., and M. Rosenbaum (2026). Mean-field equilibrium of heterogeneous agents under market impact. arXiv preprint arXiv:2609.03115.

Moitra, A. (2015). Super-resolution, extremal functions and the condition number of Vandermonde matrices. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, 821–830.

Niroula, P., S. Alkhushayni, Y. Chen, M. Bhavsar, and P. Niroula (2026). From human markets to machine markets: Transient amplification, an exact depth condition, and a maximal pre-trade guard. Working paper, Minnesota State University, Mankato.

Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer.

Wu, Y., and P. Yang (2020). Optimal estimation of Gaussian mixtures via denoised method of moments. *Annals of Statistics* 48, 1981–2007.

Zaffaroni, P. (2004). Contemporaneous aggregation of linear dynamic models in large economies.

Journal of Econometrics 120, 75–102.